1) $H(f)$ is always real valued:

$$\left.\begin{array}{l} |X(f)|^2 \in \mathbb{R} \\ |D(f)|^2 \in \mathbb{R} \end{array}\right\} H(f) \in \mathbb{R}$$

2) $D(f) \to 0 \quad \Rightarrow \quad H(f) = \frac{|X(f)|^2}{|X(f)|^2} = 1$

$D(f) \to \infty \quad \Rightarrow \quad H(f) = \frac{|X(f)|^2}{|X(f)|^2 + \infty} = 0$

$H(f) \leq 1$

$$\frac{|X(f)|^2}{|X(f)|^2 + |D(f)|^2} \leq 1 \qquad \Big| \cdot \left(|X(f)|^2 + |D(f)|^2\right)$$

$$|X(f)|^2 \leq |X(f)|^2 + |D(f)|^2 \qquad \Big| -|X(f)|^2$$

$$0 \leq |D(f)|^2 \qquad \text{always true}$$

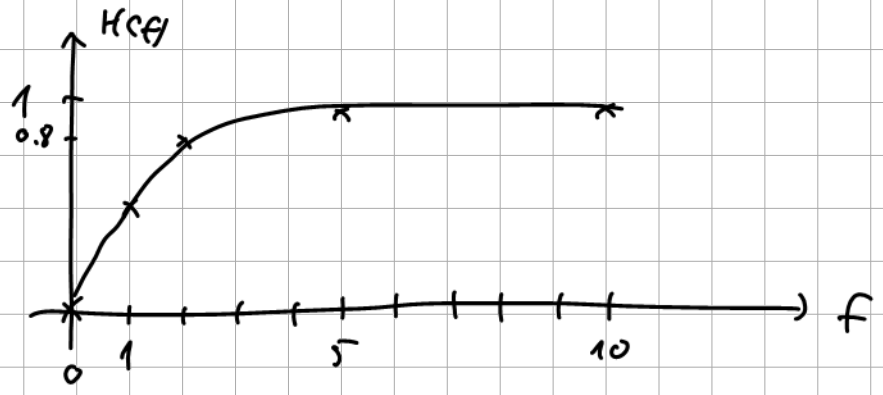
$H(f) \geq 0$

$$\frac{|X(f)|^2}{|X(f)|^2 + |D(f)|^2} \geq 0 \qquad \Big| \cdot \left(|X(f)|^2 + |D(f)|^2\right)$$

$$|X(f)|^2 \geq 0 \qquad \text{always true}$$

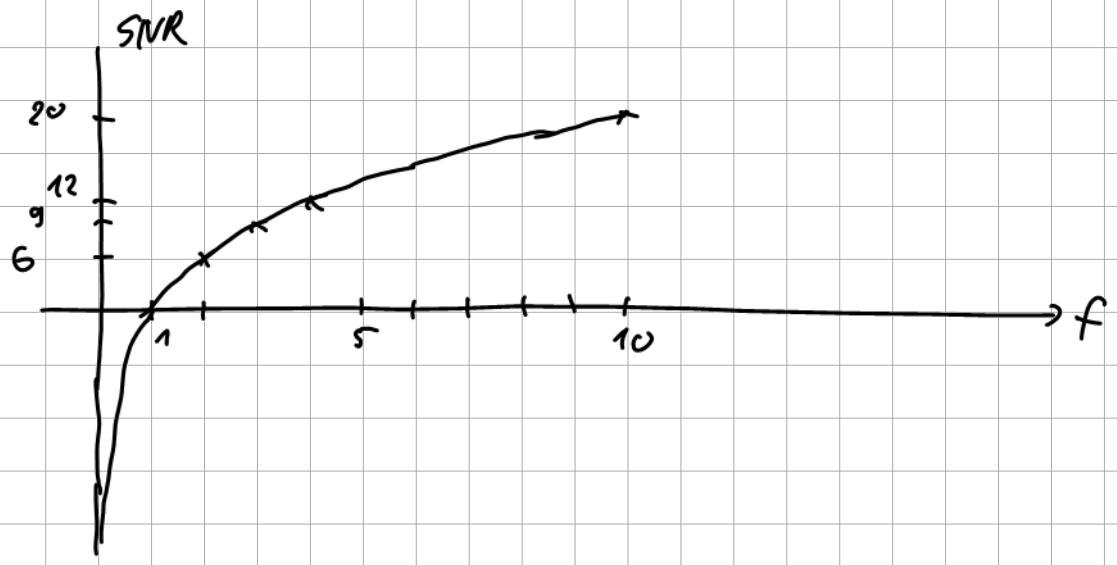
3) $X(f) = f$ $\qquad$ $|X(f)|^2 = f^2$

$$H(f) = \frac{f^2}{f^2+1}$$



4) $$SNR = 10 \log_{10} \frac{|X(f)|^2}{|D(f)|^2}$$

$$= 10 \log_{10} f^2$$



5) The Denominator is greater or equal zero.

The addition of a small positive number prevents a division by zero without affecting non-zero elements of the denominator too much.

6) Hoelder mean $= \left(\sum_n |x(n)|^p \right)^{\frac{1}{p}}$, here $p = -1$

$x(n) = (1 \quad 2 \quad 3)$

$$\Rightarrow \left(|1|^{-1} + |2|^{-1} + |3|^{-1} \right)^{\frac{1}{-1}}$$

$$= \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{3} \right)^{-1} = \underbrace{\frac{1}{1 + \frac{1}{2} + \frac{1}{3}}}_{\text{harmonic mean}} = \frac{1}{\frac{6}{6} + \frac{3}{6} + \frac{2}{6}} = \frac{6}{11}$$